

Problem Set for 28 April 2025

Important

The following exercises concern the theory of dual spaces and their associated structures.

Definition. (Kernels and Annulateurs). For convenience, let U_i denote subspaces of V , and B_i subspaces of V^* .

- $\text{ann } U := \{f \in V^* \mid f(U) = 0\}$, a subspace of V^* ;
- $\ker B := \{u \in V \mid f(u) = 0 \ (\forall f \in B)\} = \bigcap_{f \in B} \ker f$, a subspace of V .

These notions are defined for subsets in general. By the closure property:

- $\text{ann}(X) = \text{ann}(\text{span}(X))$, for $X \subseteq V$;
- $\ker(S) = \ker(\text{span}(S))$, for $S \subseteq V^*$.

Definition The 4 fundamental spaces of a matrix

- $N(A)$: null space (column vectors),
- $C(A)$: column space (column vectors),
- $L(A)$: left null space (row vectors),
- $R(A)$: row space (row vectors).

Definition For $\varphi : U \rightarrow V$, we define the dual map

$$\varphi^* : V^* \rightarrow U^*, \quad \ell \mapsto \ell \circ \varphi.$$

In other word, φ^* means the pre-composition.

Exercise 0 (Very easy exercise). Show that φ^* is injective when φ is surjective.

 Warning

U and V are not finite dimensional in general, so never take the basis.

若 $f : U \rightarrow V$ 是满的, 则对任意 W ,

$$\mathrm{Hom}_{\mathbb{F}}(V, W) \rightarrow \mathrm{Hom}_{\mathbb{F}}(U, W), \quad \alpha \mapsto \alpha \circ f$$

总是单的.

Exercise 1 Show that $\text{ann}(\text{im } \varphi)$ and $\ker(\varphi^*)$ are the same subspaces of V^* .

💡 **Tip**

You are encouraged to restate the equality $\text{ann}(\text{im } \varphi) = \ker(\varphi^*)$ in natural language. One possible (oral) answer is:

- for any $f \in V^*$ killed by the $(- \circ \varphi)$, f kills all $\varphi(u)$'s.

⚠ **Warning**

U and V are not finite dimensional in general, so never take the basis.

$$[f \in \text{ann}(\text{im } \varphi)] \iff [f(\text{im } \varphi) = 0] \iff [f \circ \varphi = 0] \iff [f \in \ker(\varphi^*)]$$

Exercise 2 When φ is surjective, show that $\text{im}(\varphi^*)$ and $\text{ann}(\ker \varphi)$ are the same subspaces of U^*

💡 **Tip**

You are encouraged to restate the equality $\text{im}(\varphi^*) = \text{ann}(\ker \varphi)$ in natural language. One possible (oral) answer is:

- for any $g \in U^*$ killing $\ker \varphi$, g factors through φ .

⚠ **Warning**

U and V are not finite dimensional in general, so never take the basis.

$$[f \in \text{ann}(\ker \varphi)] \iff [(\ker \varphi \xrightarrow{\iota} U \xrightarrow{f} \mathbb{F}) = 0] \iff [f \text{ 通过 } \text{cok}(\iota) \text{ 分解}] \iff [f \in \text{im}(\varphi^*)].$$

Exercise 3 When $\varphi : S \hookrightarrow V$ is injective, show that both $(V/S)^*$ and $\text{ann}(S)$ are isomorphic to $\ker(\varphi^*)$.

⚠ **Caution**

Can we write $(V/S)^* = \text{ann}(S)$ here; consider the appropriate interpretation involving isomorphisms.

⚠ **Warning**

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由 **Ex 1**, $\text{ann}(S)$ 与 $\ker(\varphi^*)$ 是相同的 V^* 的子空间. 下考虑单射

$$(- \circ \pi) : (V/S)^* \rightarrow V^*, \quad l \mapsto l \circ \pi.$$

这一单射的像是形如 $V \rightarrow \mathbb{F}$ 的且被商空间 V/S 分解的映射, 从而恰好是将 S 零化的映射.

综上, $(V/S)^* \simeq \text{ann}(S) = \ker(\varphi^*)$, 同构的实现方式是右复合满射 π .

Exercise 4 For any V , we define the evaluation as before:

$$\Phi_V : V \rightarrow V^{**}, \quad v \mapsto \begin{bmatrix} V^* & \rightarrow & \mathbb{F} \\ \ell & \mapsto & \ell(v) \end{bmatrix}.$$

We define $\varphi^{**} := (\varphi^*)^*$, that is, the pre-composition of $\ell : U^* \rightarrow \mathbb{F}$ by $\varphi^* : V^* \rightarrow U^*$. Show the equality of the compositions

$$\left[U \xrightarrow{f} V \xrightarrow{\Phi_V} V^{**} \right] = \left[U \xrightarrow{\Phi_U} U^{**} \xrightarrow{f^{**}} V^{**} \right].$$

In other words, $\Phi_V(f(u)) = f^{**}(\Phi_U(u))$ for any $u \in U$. This is why we say Φ is natural.

💡 Tip

This can be demonstrated using string manipulations, even if the conceptual meaning is not yet fully clear.

⚠ Warning

U and V are not finite dimensional in general, so never take the basis.

一方面, $\left[U \xrightarrow{f} V \xrightarrow{\Phi_V} V^{**} \right]$ 描述作

$$U \rightarrow V^{**}, \quad u \mapsto \begin{bmatrix} V^* & \xrightarrow{\Phi_V(f(u))} & \mathbb{F} \\ \ell & \mapsto & \ell(f(u)) \end{bmatrix}$$

另一方面, $\left[U \xrightarrow{\Phi_U} U^{**} \xrightarrow{f^{**}} V^{**} \right]$ 描述作

$$U \rightarrow V^{**}, \quad u \mapsto \begin{bmatrix} V^* & \xrightarrow{f^{**}(\Phi_U(u))} & \mathbb{F} \\ \ell & \mapsto & (f^{**}(\Phi_U))(\ell) \end{bmatrix}$$

此处

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$$(f^{\ast}(\Phi_U(u)))(\ell) = ((\Phi_U(u)) \circ f^{\ast})(\ell) = (\Phi_U(u))(f^{\ast}(\ell)) = (\Phi_U(u))(\ell \circ f) = (\ell \circ f)(u).$$

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第一部分的操作是 $[(fu)^{\ast}] \circ f$, 第二部分的操作是 $[(f^{\ast}u)^{\ast}] \circ f$. 以上运算默认右结合, 结论是 $(fu)^{\ast} = f^{\ast}u^{\ast}$.

Exercise 5 Show that

$$a : \mathbb{F}^{m \times n} \rightarrow (\mathbb{F}^{n \times m})^*, \quad M \mapsto \begin{bmatrix} \mathbb{F}^{n \times m} & \rightarrow & \mathbb{F} \\ X & \mapsto & \text{trace}(M \cdot X) \end{bmatrix}$$

is a linear map. And show that a is surjective.

💡 **Tip**

Consider how to demonstrate the surjectivity efficiently?

线性性的检验略. 若 $\dim U = \dim V$, 则 $f : U \rightarrow V$ 单等价于满等价于同构. 为证明 a 满, 只需证明 a 单.

选定非零的 M , 例如 $m_{i,j} \neq 0$, 往证 $a(M)$ 非零即可. 显然

$$(a(M))E_{j,i} = \text{tr}(ME_{j,i}) = m_{i,j} \neq 0.$$

Exercise 6 (For recreation). For finite dimensional vector space V over arbitrary field $\mathbb{F}$, we write

- $V := \mathbb{F}^{n \times 1}$, the space of columns vectors with common basis $\{e_i\}_{i=1}^n$, and
- $V^* := \mathbb{F}^{1 \times n}$, the space of row vectors with common basis $\{f^j\}_{j=1}^n$,

Define a **non-degenerate bilinear pairing** via the standard row-column multiplication:

$$V^* \times V \rightarrow \mathbb{F}, \quad (u_{\text{row}}; v_{\text{column}}) \mapsto u \cdot v.$$

Using this pairing, observe:

1. The left multiplication of a matrix $(A \cdot) : V \rightarrow V$, induces a dual linear map of the right multiplication of the matrix $(\cdot A) : V^* \rightarrow V^*$. Remind the identity

$$(u; A \cdot v) = (u \cdot A; v).$$

2. Once V^* is viewed as a column space via the transpose $(-)^T : V^* \rightarrow V$, then:

$$(\cdot A) = V^* \xrightarrow{(-)^T} V \xrightarrow{A^T} V \xrightarrow{(-)^T} V^*.$$

Hence, if $\varphi : U \rightarrow V$ has the matrix form $M \in \mathbb{F}^{m \times n}$, φ^* has the matrix form M^T . Now $(M^T)^T = M$ is consistent with the natural isomorphism.

The question: explain $\ker \varphi = \ker(\text{im}(\varphi^*))$ by

$$N(A) = \ker(A \cdot) = \bigcap_{u \in V^*} \ker(uA \cdot) = \ker(\text{im}((A \cdot)^*)) = (L(A^T))^T.$$

Analogously, explain $\text{FD} \times \{Zi\}_{i=1}^6$ in [this exercise](#).

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